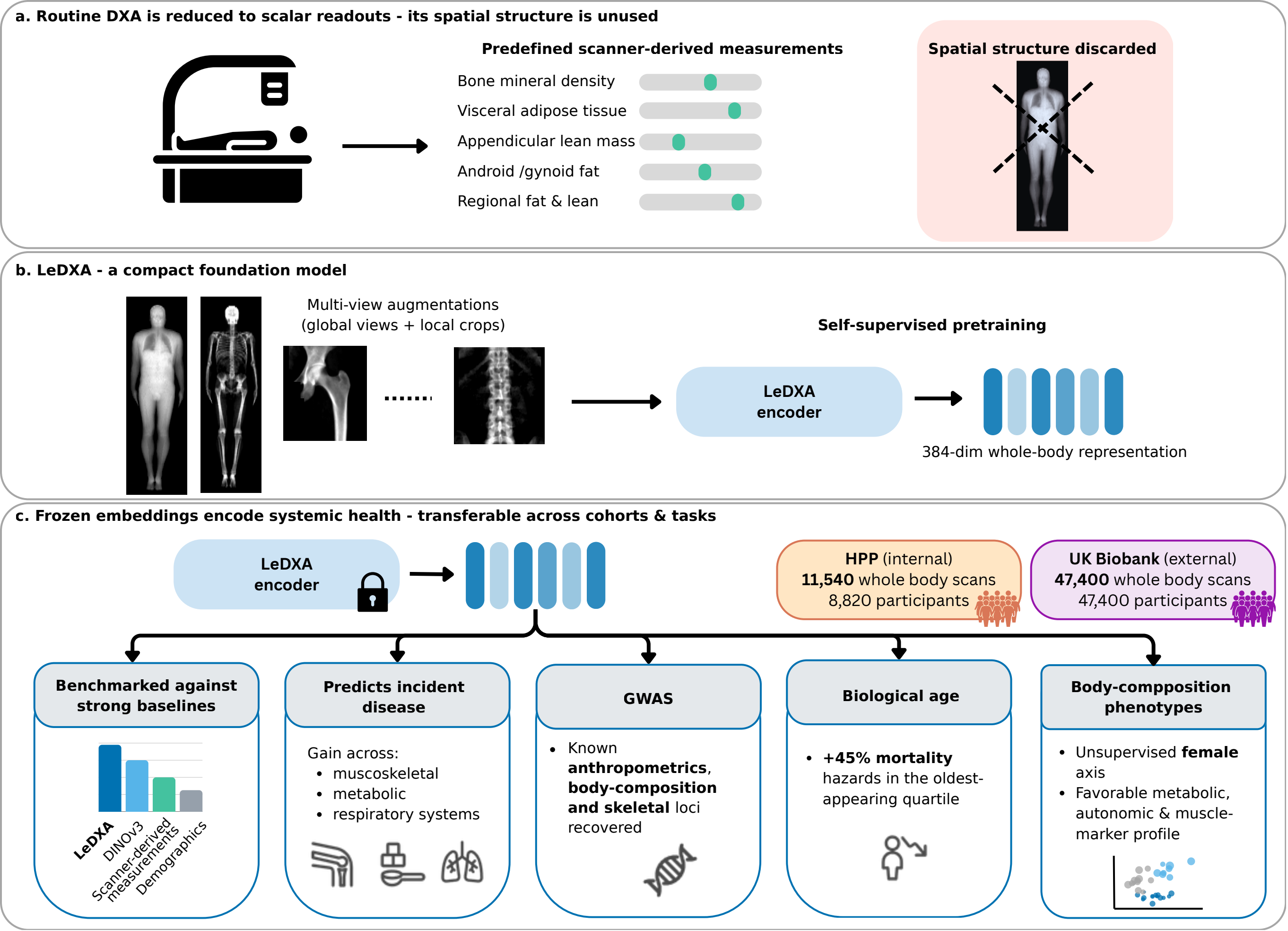
a. Routine DXA is reduced to scalar readouts - its spatial structure is unused
Predefined scanner-derived measurements
Bone mineral density
Visceral adipose tissue
Appendicular lean mass
Android /gynoid fat
Regional fat & lean
Spatial structure discarded
b. LeDXA - a compact foundation model
Multi-view augmentations
(global views + local crops)
Self-supervised pretraining
LeDXA
encoder
384-dim whole-body representation
c. Frozen embeddings encode systemic health - transferable across cohorts & tasks
LeDXA
encoder
HPP (internal)
11,540 whole body scans
8,820 participants
UK Biobank (external)
47,400 whole body scans
47,400 participants
Benchmarked against strong baselines
LeDXA
DINOv3
Scanner-derived measurements
Demographics
Predicts incident disease
Gain across:
muscoskeletal
metabolic
respiratory systems
GWAS
Known anthropometrics, body-composition and skeletal loci recovered
Biological age
+45% mortality hazards in the oldest-appearing quartile
Body-compposition phenotypes
Unsupervised female axis
Favorable metabolic, autonomic & muscle-marker profile